

Al-Aqsa University

Faculty of Computers and IT

Engineering and Networks Dept.



جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم الهندسة والشبكات

Operating Systems for Intelligent Systems (ENGS2306)

Lecture 8: Linux Networking Basics

Lecturer: Eng. Mohammed E. Alhams

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Lecture 8: Linux Networking Basics

Today's Objectives:

1. Understand network interfaces and how to view their configuration.
 2. Learn to test network connectivity using ping.
 3. Use ss and netstat to view active network connections and listening services.
 4. Learn to securely access and manage remote Linux systems using ssh.
 5. Securely transfer files over the network using scp.
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Recap: From Automation to Communication

In **Lecture 7**, we mastered advanced scripting with functions, arrays, and error handling. We learned how to write robust scripts to manage a system efficiently.

The Connection to Today:

What makes an intelligent system "intelligent"? Often, it's the ability to **communicate**—with other devices, with servers, or with users.

- The scripts we write often need to manage network services or run on remote machines.
- To manage a fleet of IoT devices, you don't walk up to each one. You connect to them over the network.

Today, we learn the fundamental networking commands that make this communication possible.

Viewing Network Configuration: ip addr

Every device on a network needs an address. The ip command is the modern tool for managing network configuration in Linux.

Command: ip addr show (or ip a for short)

This command lists all network interfaces on your system and their IP addresses.

Example Output:

```
1: lo: <LOOPBACK,UP,...>
   inet 127.0.0.1/8 scope host lo
2: ens33: <BROADCAST,MULTICAST,UP,...>
   link/ether 00:0c:29:b1:c8:f3
   inet 192.168.75.128/24 brd 192.168.75.255 scope global ...
```

- **lo (Loopback):** The "localhost" interface. Its IP address is always 127.0.0.1. It's used for services on your own machine to talk to each other.

- **ens33 (or eth0):** Your physical/virtual Ethernet card.
 - **inet 192.168.75.128/24:** The IPv4 address of your machine on the local network. This is how other devices reach you.
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Testing Connectivity: ping

The ping command is the first tool you should use to check if a device is online and reachable over the network. It sends a small packet (an ICMP echo request) and waits for a reply.

Syntax: ping <hostname_or_IP>

Examples:

```
# Ping Google's public DNS server to check internet connectivity
ping -c 4 8.8.8.8
```

```
# Ping another device on your local network
ping 192.168.1.101
```

```
# Ping your own machine to check if the network stack is working
ping localhost
```

- -c 4 tells ping to send only 4 packets and then stop.
 - A successful reply means the connection is alive. "Destination Host Unreachable" or 100% packet loss means there's a problem.
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Viewing Connections: ss and netstat

How do you see what services are "listening" for connections on your machine?

- ss (socket statistics) is the modern, faster command.
- netstat is the older, legacy command (but still widely used).

Most Useful Command: ss -tulpn

- -t: Show TCP sockets.
- -u: Show UDP sockets.
- -l: Show only listening sockets.
- -p: Show the process using the socket.
- -n: Show numerical addresses (don't resolve hostnames).

Example Output:

State	Recv-Q	Send-Q	Local Address:Port	Peer Address:Port	Process
LISTEN	0	4096	127.0.0.1:631	0.0.0.0:*	users:
(("cupsd",pid=839,fd=7))					
LISTEN	0	128	0.0.0.0:22	0.0.0.0:*	users:
(("sshd",pid=1023,fd=3))					

This output shows that the CUPS printing service is listening on port 631 and the SSH service is listening on port 22.

Secure Remote Management: ssh

SSH (Secure Shell) is the most important tool for a Linux administrator. It provides an encrypted, secure command-line login to a remote machine.

Syntax: `ssh <username>@<hostname_or_IP>`

For Intelligent Systems: This is how you manage "headless" devices—systems without a monitor or keyboard, like a Raspberry Pi in a weather station or a server in a data center.

Example:

```
# Log in to a remote server with the user 'admin'
ssh admin@192.168.1.200

# After entering the password, you get a shell on the remote machine.
# Every command you type now runs on that machine.
admin@192.168.1.200:~$ hostname
server-alpha
admin@192.168.1.200:~$ exit
logout
Connection to 192.168.1.200 closed.
```

Secure File Transfer: scp

SCP (Secure Copy Protocol) uses SSH to securely copy files and directories between two locations on a network. It works just like the `cp` command, but with a remote destination.

Syntax:

`scp [options] <source> <destination>`

Examples:

```
# Copy a local file to a remote server
scp ./my_script.sh admin@192.168.1.200:/home/admin/scripts/

# Copy a file from a remote server to your local machine
scp admin@192.168.1.200:/var/log/syslog ./remote_logs/
```

This is essential for deploying new code to an IoT device or retrieving log files from it for analysis.

Practical Scenario: Local Network Loop

Let's test these tools on our own machine. This confirms they are working without needing a second computer.

1. Open your terminal.
 2. Find your primary IP address: `ip a`. Note it down.
 3. Check internet connectivity: `ping -c 3 8.8.8.8`.
 4. Check if your own network stack is up: `ping -c 3 localhost`.
 5. Find out which services are listening: `ss -ltn`. Look for port 22 (SSH).
 6. Connect to your own machine using SSH: `ssh your_username@localhost`. (Replace `your_username`).
 7. You will be asked to trust the key fingerprint (type yes) and then enter your password. You are now in a new shell session on your own machine.
 8. Create a test file: `touch ssh_test.txt`. Then type `exit` to close the SSH session.
 9. Securely copy that file to your home directory's Downloads folder using `scp`.
`scp localhost:/home/your_username/ssh_test.txt ~/Downloads/`
 10. Verify the file was copied: `ls ~/Downloads/`.
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The Idea Behind the Assignment

Why are networking skills non-negotiable for Intelligent Systems?

- **Remote Management:** An intelligent system is rarely used in isolation. An IoT sensor array, a smart factory robot, or an autonomous vehicle must be managed, updated, and debugged remotely. `ssh` is the key to this.
- **Data Transfer:** Intelligent systems generate or consume data. A smart camera needs to send video feeds, and a machine learning model needs to receive updates. `scp` (and other tools like `rsync`) are used for these secure data transfers.
- **Troubleshooting:** When an IoT device goes offline, you don't just assume it's broken. The first step is to ping it. If it doesn't respond, you check the router. If it responds, but the service isn't working, you `ssh` into it and use `ss` to see if the application is listening on the correct port. This logical process of troubleshooting is a core engineering skill.

Assignment 7: Network & Connectivity Audit (12%)

Objective: To use core networking tools to document and test your system's connectivity and securely interact with it.

Tasks:

1. Use `ip a` to find the IPv4 address of your main network interface (e.g., `ens33` or `eth0`).
2. Use `ping -c 4 8.8.8.8` to test and show your internet connectivity.
3. Use `ping -c 4 localhost` to test and show your local network stack is working.
4. Use `ss -tulpn` to list all listening TCP and UDP services. Identify the line for the SSH service (port 22).
5. Securely connect to your own machine with `ssh your_username@localhost`. Once logged in, run the `whoami` command.
6. **Verification:** Capture the output of all commands from steps 1-5 in your report.
7. **Report:** Write a brief paragraph (3-5 sentences) explaining why a service like SSH "listens" on a specific port. What is the purpose of port 22 in this context, and why is it important that only the correct service is listening on it?

Assignment Submission Instructions

- **File Name:** Name your submission file exactly as follows:
studentnum-studentname-Assignment7.pdf
 - *Example:* 13012070xx-MohammedEmadAlhams-Assignment7.pdf
- **Terminal Prompt:** Your name must appear in the terminal prompt for your screenshots and command outputs.
- **File Format:** Your submission must be a **single PDF file** containing:
 - The outputs for each command in the assignment tasks.
 - Your final report paragraph explaining the purpose of a listening port.
- **Academic Integrity:**
 - **Warning:** Identical submissions will be considered cheating and will result in a grade of **ZERO**.
 - **Warning:** AI-generated solutions or reports will result in a grade of **ZERO**.

Next Steps & Resources

Practice This Week:

- Explore the options for ping and ssh. Can you change the interval of a ping? Can you run a single command over SSH without a full login (ssh user@host "uptime")?
- If you have a Raspberry Pi or another Linux machine, practice ssh and scp between them.

Next Week: Lecture 9 - Midterm Practical Review

- We will pause to consolidate everything we have learned in the first half of the course. This will be a critical review session covering commands, scripting, and networking to prepare you for the midterm assessment.

Gaza Connectivity Tips:

- The ssh, ping, and scp commands can all be tested using localhost without needing an internet connection, which is great for offline practice.
- Find a stable time to download any resources.

Resources:

- Introduction to Linux Networking: <https://www.digitalocean.com/community/tutorials/an-introduction-to-linux-networking>
- SSH Essentials: <https://www.hostinger.com/tutorials/ssh-essentials>
- Read the manual pages: man ip, man ping, man ssh, man scp
- Course Materials on Moodle